

Example Runs:

Input 1:

r Copy Edit

Enter an arithmetic expression: $(3 + 5) * 2$

Output 1:

makefile Copy Edit

Result: 16

Input 2:

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Enter an arithmetic expression: $10 + 2 * 3$

Output 2:

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Result: 16

(Multiplication has higher precedence than addition, so $2 * 3$ is evaluated first, then added to 10.)

Input 3:

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Enter an arithmetic expression: $(10 + 2) / 4$

Output 3:

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Result: 3

(Parentheses force $10 + 2$ to be evaluated first, then divided by 4.)

Input 4:

```
r
```

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Enter an arithmetic expression: `50 - 10 / 2`

Output 4:

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Result: 45

(Division is evaluated first: `10 / 2 = 5`, then subtracted from 50.)

Input 5:

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r
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Enter an arithmetic expression: `5 * (2 + 3) / 5`

Output 5:

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Result: 5

(Parentheses ensure `2 + 3 = 5`, then `5 * 5 = 25`, finally `25 / 5 = 5`.)